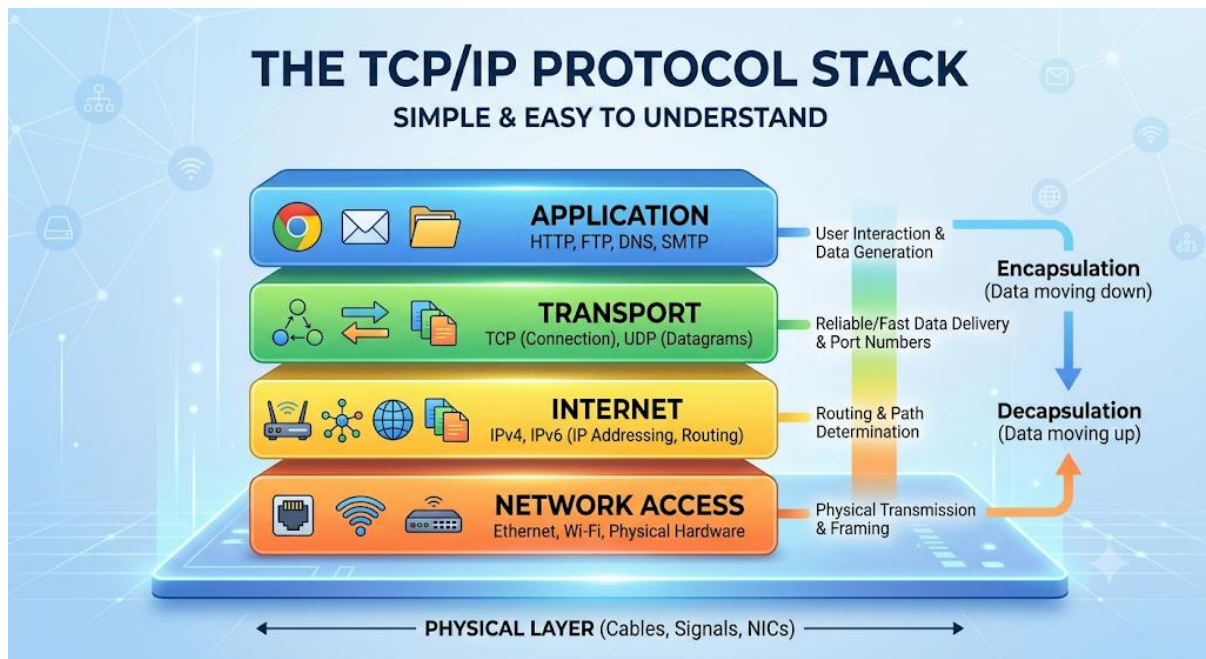


How the Internet Delivers a Web Page

Website Chosen is www.wikipedia.org

1. TCP/IP Protocol Stack



Application Layer: Provides services such as HTTP and DNS for web communication.

Transport Layer: Uses TCP to ensure reliable and error-free data transmission.

Internet Layer: Routes packets between the client and the server using IP addresses.

Network Access Layer: Transfers data over physical networks such as Wi-Fi or Ethernet.

2. Role of DNS

When a user enters www.wikipedia.org in a web browser, the browser sends a DNS request. The DNS server translates the domain name into an IP address. Using this IP address, the browser locates the Wikipedia web server and establishes communication.

3. HTTP Communication

After obtaining the IP address, the browser sends an **HTTP GET** request to the Wikipedia server. The server processes the request and sends back an HTTP response containing the webpage's HTML, CSS, images, and other resources. The browser then displays the webpage to the user. Secure communication is provided through **HTTPS**.

4. TCP Reliability

TCP provides reliable communication by:

- Establishing a connection using the Three-Way Handshake.
- Dividing data into segments.
- Sending acknowledgements (ACKs).
- Retransmitting lost packets.
- Delivering data in the correct order.

This ensures that the Wikipedia webpage is loaded correctly without missing information.

5. Architecture

Wikipedia follows a **Client–Server Architecture**.

- **Client:** User's web browser.
- **Server:** Wikipedia web server.

The browser sends requests, and the server responds by delivering the requested webpage.

6. HTTP vs FTP

HTTP	FTP
Used to access web pages	Used to transfer files

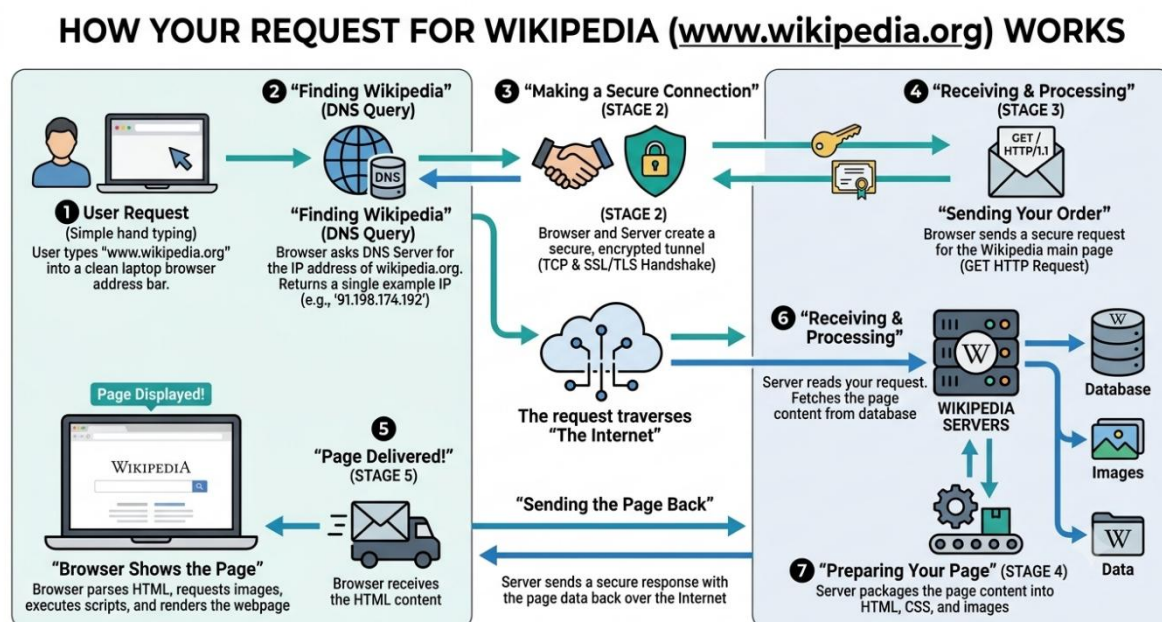
Uses Port 80 (HTTPS: 443)	Uses Ports 20 and 21
Stateless protocol	Stateful protocol
Used by web browsers	Used by FTP clients

7. Cookies

Cookies store small amounts of information in the user's browser to improve the browsing experience.

Example: If you select a preferred language on Wikipedia, cookies remember your choice and automatically display pages in that language during future visits.

8. Flow Diagram



9. Conclusion

This assignment explains how the Internet delivers a webpage using www.wikipedia.org as an example. It demonstrates the roles of the TCP/IP protocol stack, DNS, HTTP, TCP, cookies, and the client-server architecture in web communication. DNS converts the website name into an IP address, HTTP requests and receives webpage data, and TCP ensures reliable and

ordered delivery of information. Cookies enhance the user experience by remembering preferences and settings. Together, these technologies enable users to access websites quickly, securely, and efficiently.

Understanding these protocols provides a clear insight into the functioning of the Internet and highlights their importance in everyday activities such as learning, browsing information, communication, and accessing online services